PUBP 233
INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

Week 7: The great escape, Age of Pandemics

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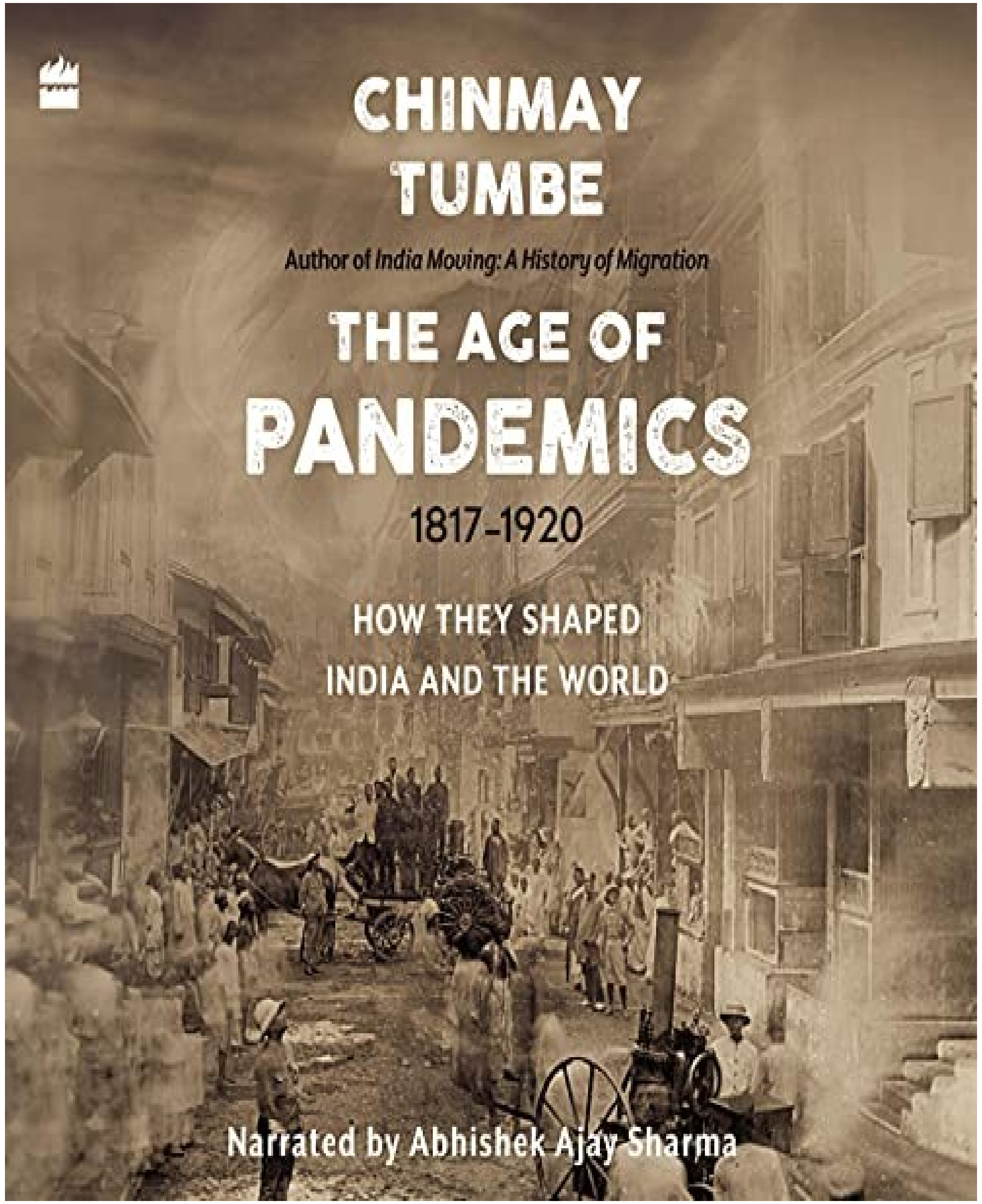
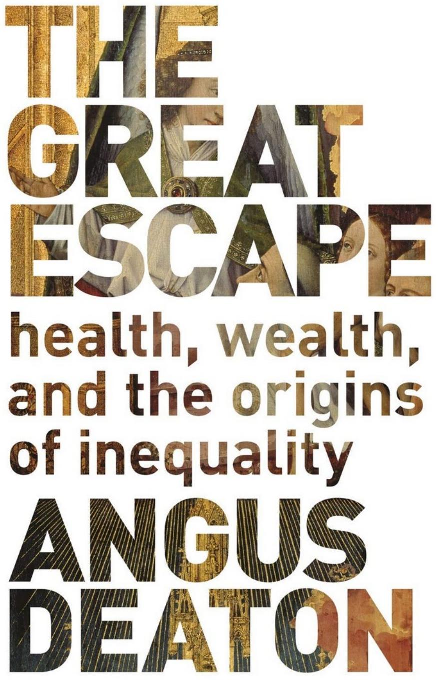
THE GREAT ESCAPE



"***Once upon a time, we barely lived before we died. We would celebrate on average only twenty-five birthdays in our lifetimes and we rarely grew old enough to see our grandchildren.*** *Mass mortality through war, famine, natural disasters and epidemics was a way of life. We began to live longer only when we fought fewer wars, understood how to prevent famines, grew resilient to natural disasters, and learnt how to control the spread of disease."*

Excerpt From: Chinmay Tumbe. "Age Of Pandemics (1817-1920): How they shaped India and the World." iBooks.



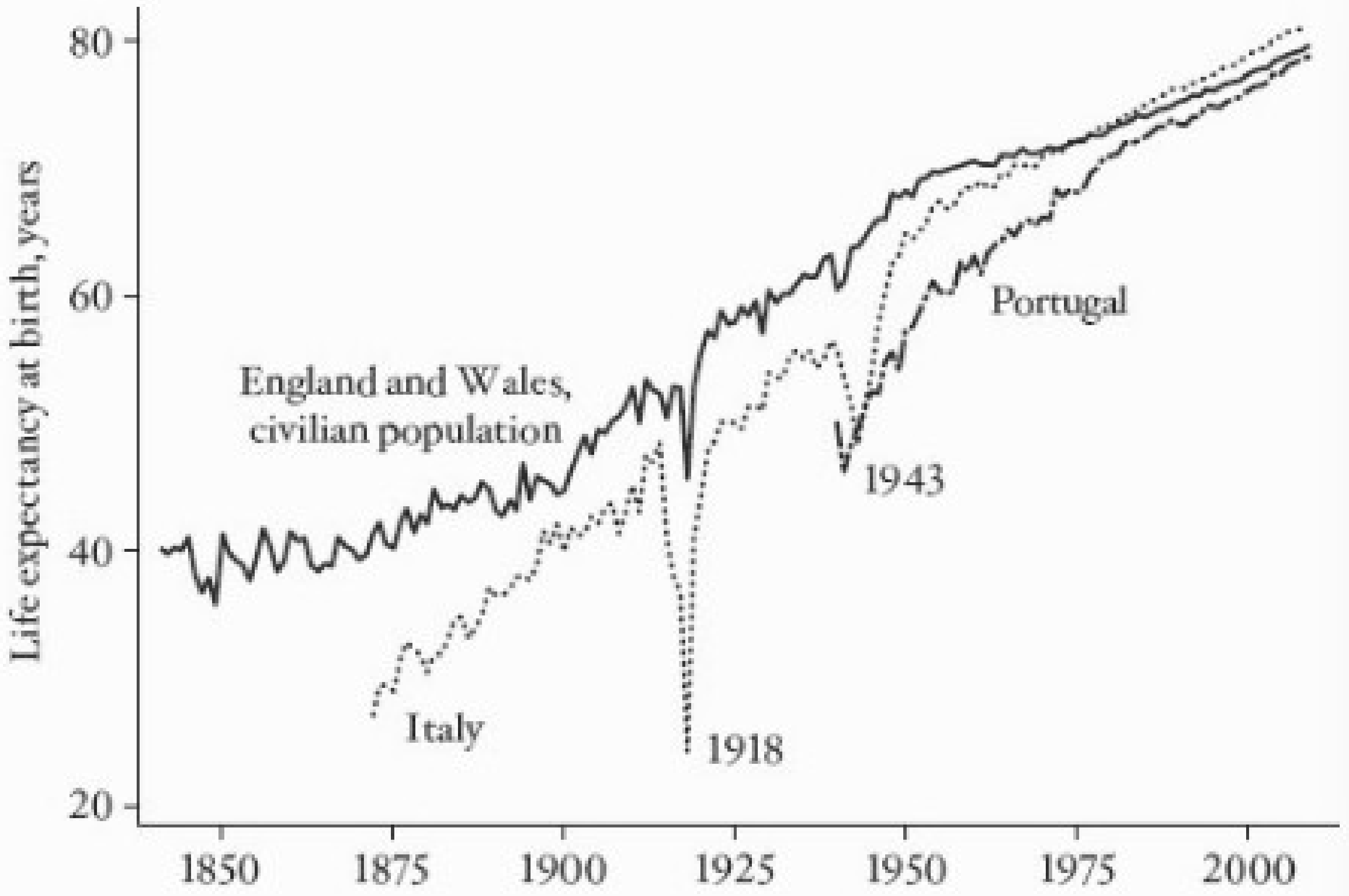


FIGURE 4 Life expectancy from 1850: England and Wales, Italy, and Portugal.

LIFE AND DEATH

HEALTH– OVER TIME AND DEVELOPED COUNTRIES

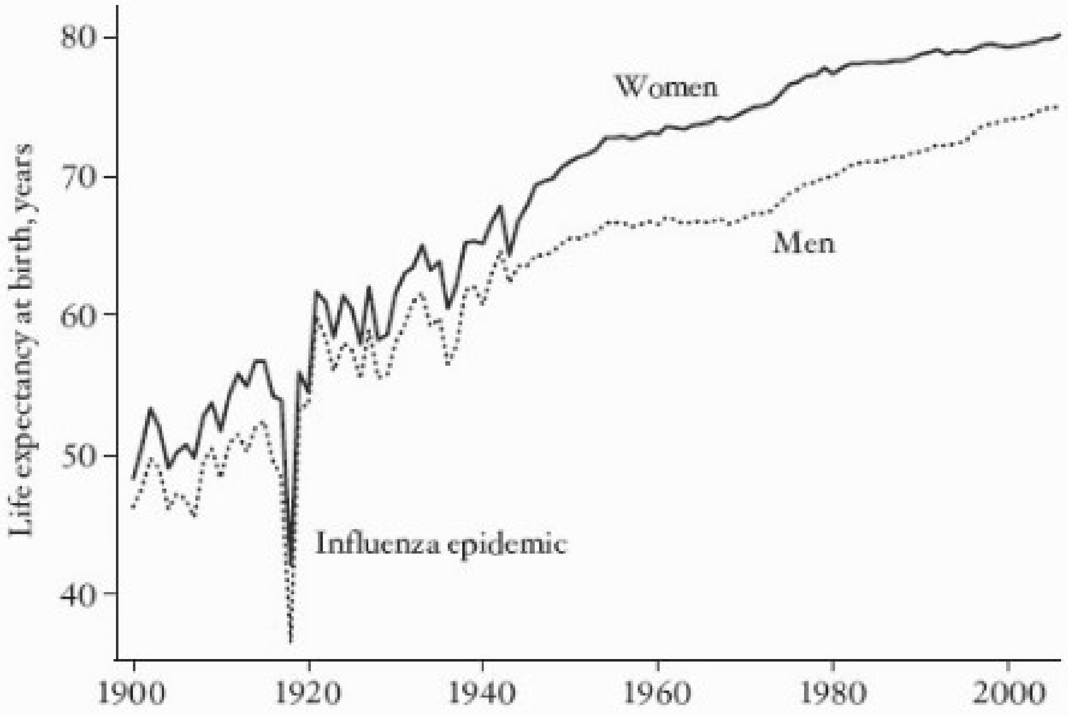


FIGURE 1 Life expectancy for men and women in the United States.

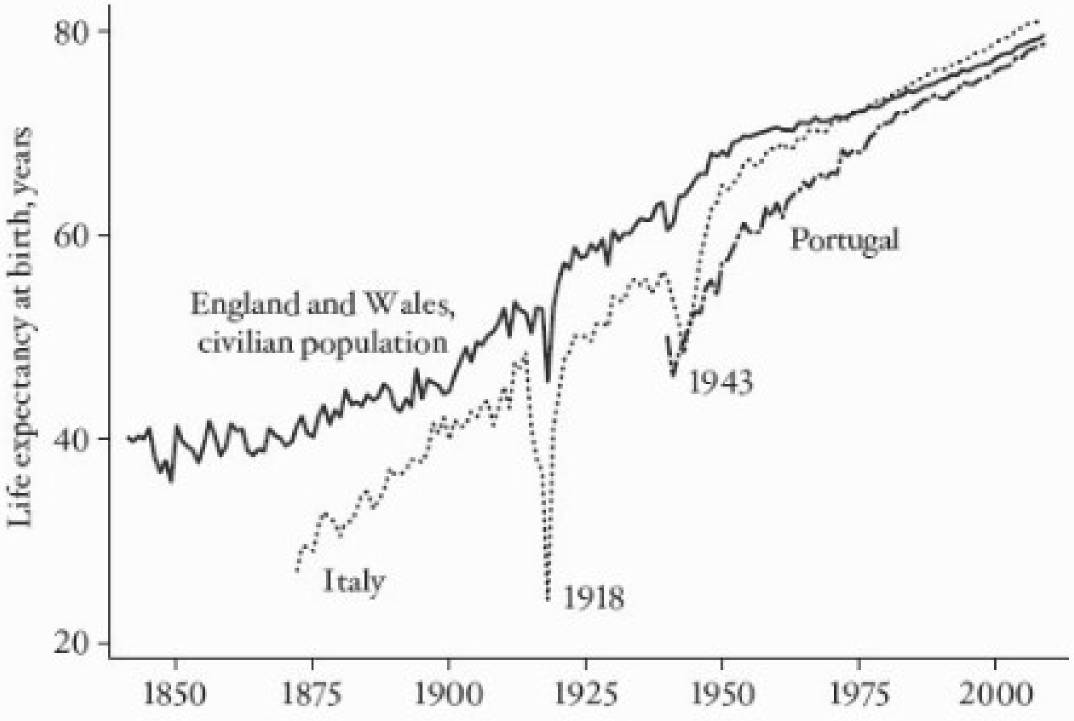


FIGURE 4 Life expectancy from 1850: England and Wales, Italy, and Portugal.

QUESTIONS

What caused life expectancy to double, from 40 years to almost 80 years, over a century and a half?

“Given the long history of thousands of years of stable or even falling life expectancy, this is surely one of the most dramatic, rapid, and favorable changes in human history. Not only will almost all newborns live to be adults, but each young adult has more time to develop his or her skills, passions, and life, a huge increase in capabilities and in the potential for wellbeing. ”

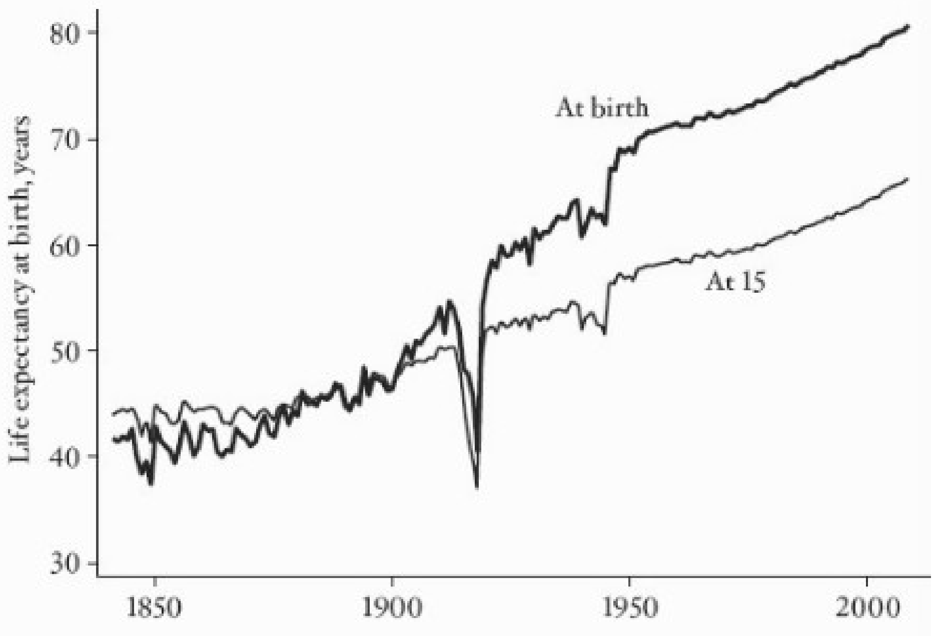


FIGURE 5 Life expectancy at birth and at 15: England and Wales, total population.

“is that, until around 1900, adult life expectancy in Britain was actually higher than life expectancy at birth. In spite of having lived for 15 years, these teenagers could expect a longer future than when they were born.”

Excerpt From: Deaton, Angus. “The Great Escape.” iBooks.

MORTALITY CHANGE OVER TIME

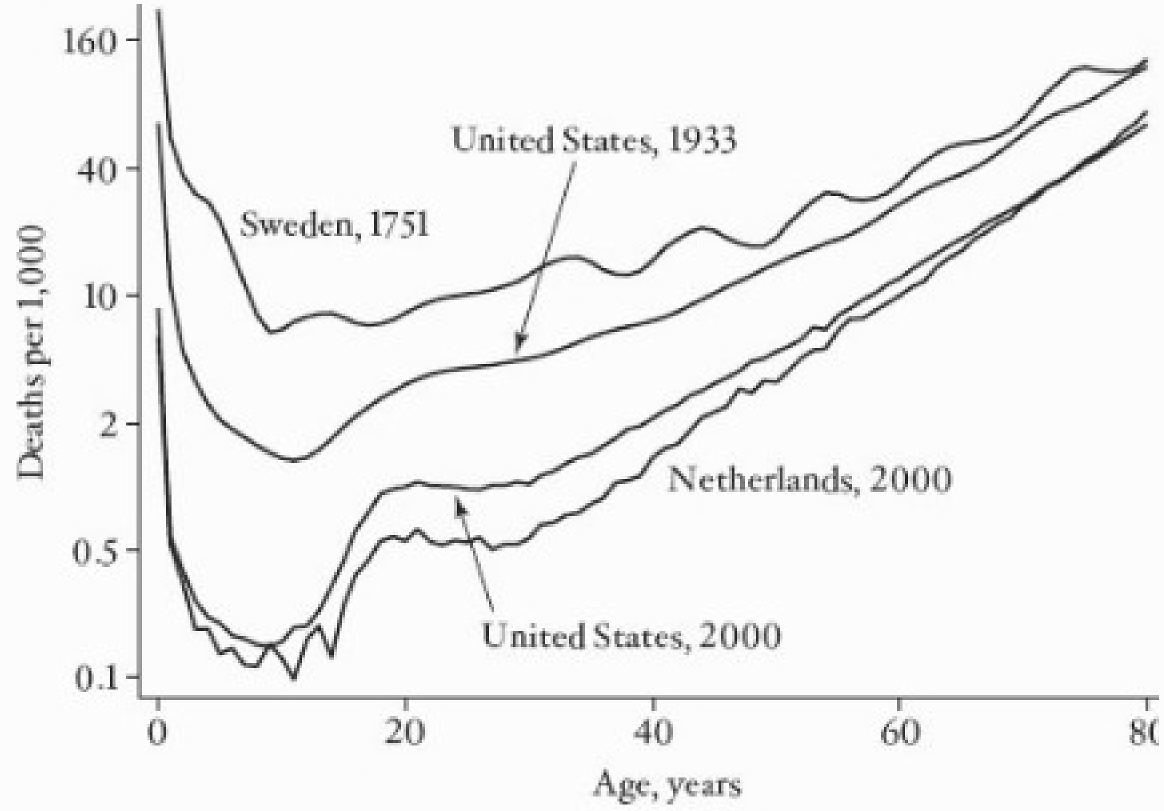


FIGURE 2 Mortality rates by age, selected countries and periods.

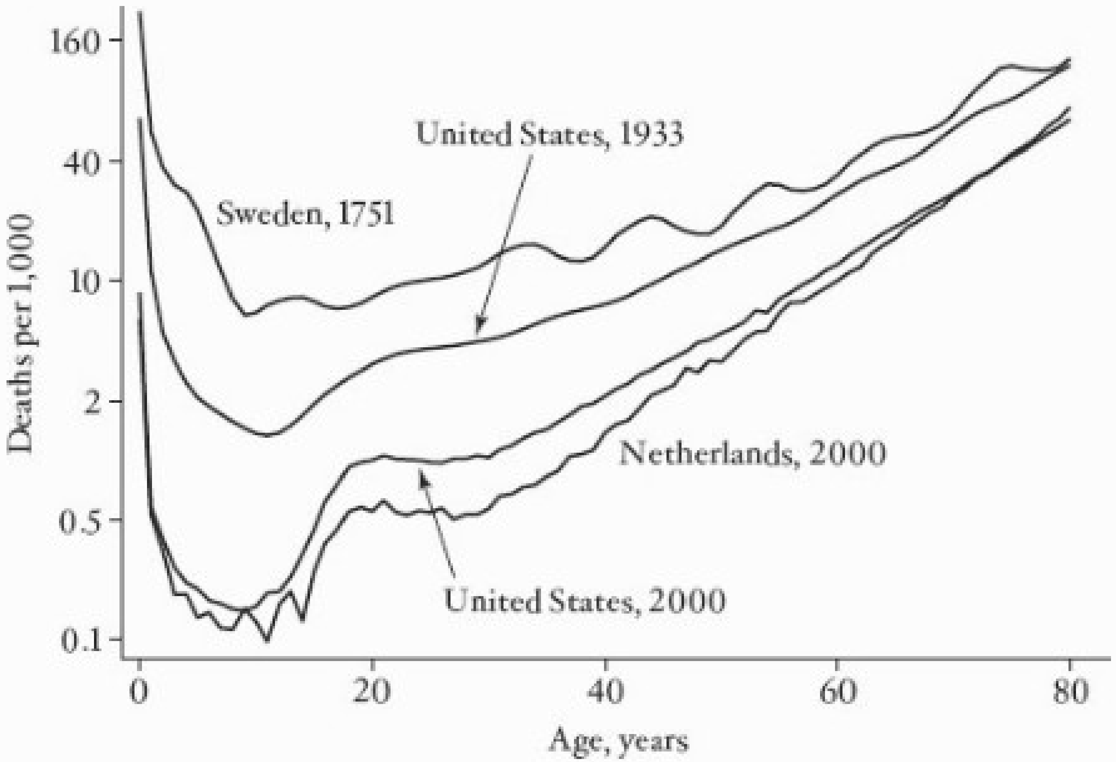


FIGURE 2 Mortality rates by age, selected countries and periods.

NUTRITION, SANITATION-PREVENTION OF DEATHS FROM EARLY CHILDHOOD DISEASES

TROPICAL COUNTRIES

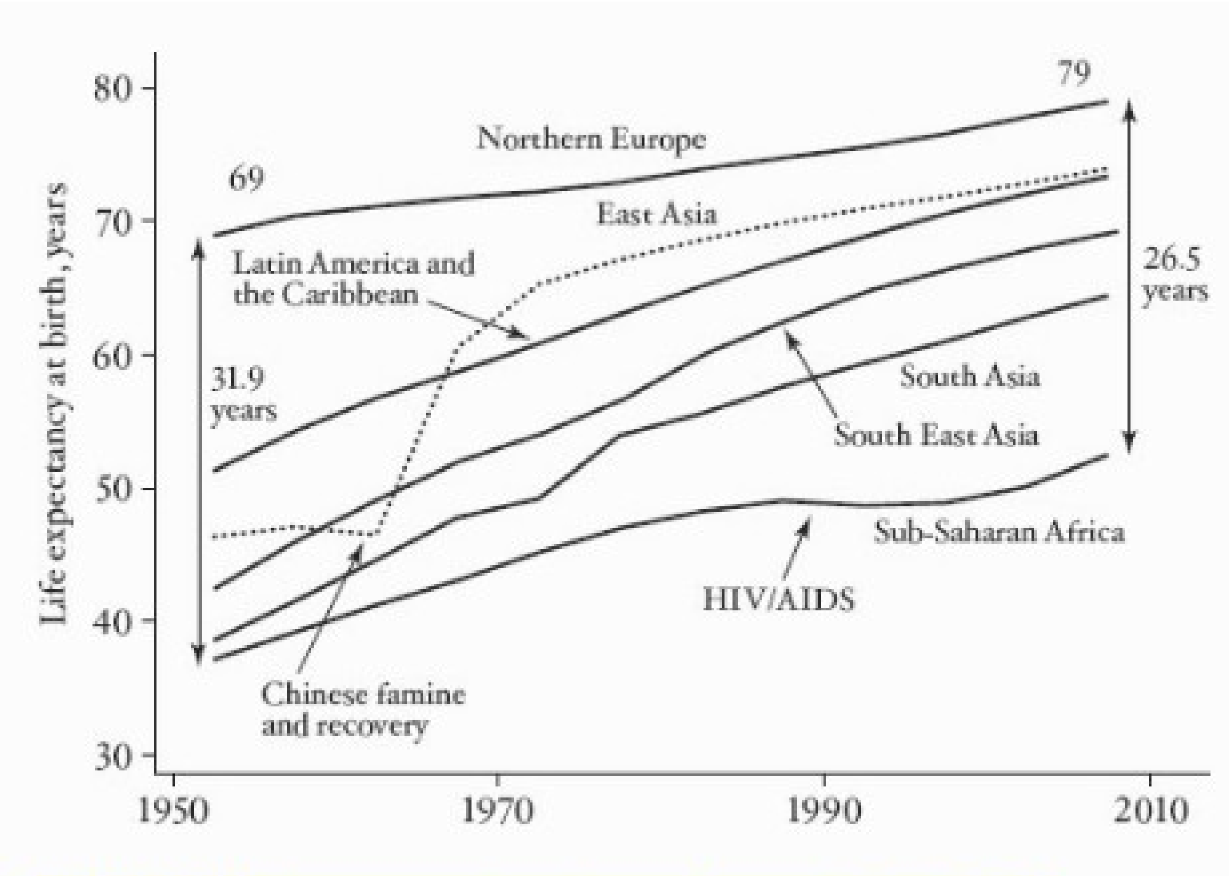


FIGURE 1 Life expectancy in regions of the world since 1950.

ESCAPING DEATH IN TROPICS

In 1850, the germ theory had yet to be established. By 1950, it was common knowledge,…That India today has higher life expectancy than Scotland in 1945—in spite of a per capita income that Britai

There are many countries where large fractions of children still die, and there are three dozen countries where more than 10 percent die before their fifth birthday. They are not dying of the "new" diseases, like HIV/AIDS, or exotic tropical diseases for which there is no cure. They are dying from the same diseases that killed European children in the seventeenth and eighteenth centuries, intestinal and respiratory infections and malaria, most of which we have known how to treat for a long time. These children are dying from the accident of where they were born, and they would not be dying had they been born in Britain, Canada, France, or Japan.

What is it that maintains these inequalities? What is it that makes it so dangerous to be born in Ethiopia or Mali or Nepal, and so safe to be born in Iceland or Japan or Singapore? Even in a country like India, where mortality rates have fallen rapidly, large fractions of children remain malnourished; they are skinnier and shorter than they ought to be for their age, and their parents are among the shortest adults on the planet, perhaps even shorter than the stunted adults in eighteenth-century England. Even today, and in spite of India being one of the fastest-growing countries in the world, why is it that so many Indians are trapped in the destitution that was the ultimate outcome of the Neolithic revolution?

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The most rapid increases in life expectancy came very soon after the war. The demographer Davidson Gwatkin reports that around 1950, **countries such as Jamaica, Malaysia, Mauritius, and Sri Lanka saw annual increases in life expectancy of more than one year for more than a decade**

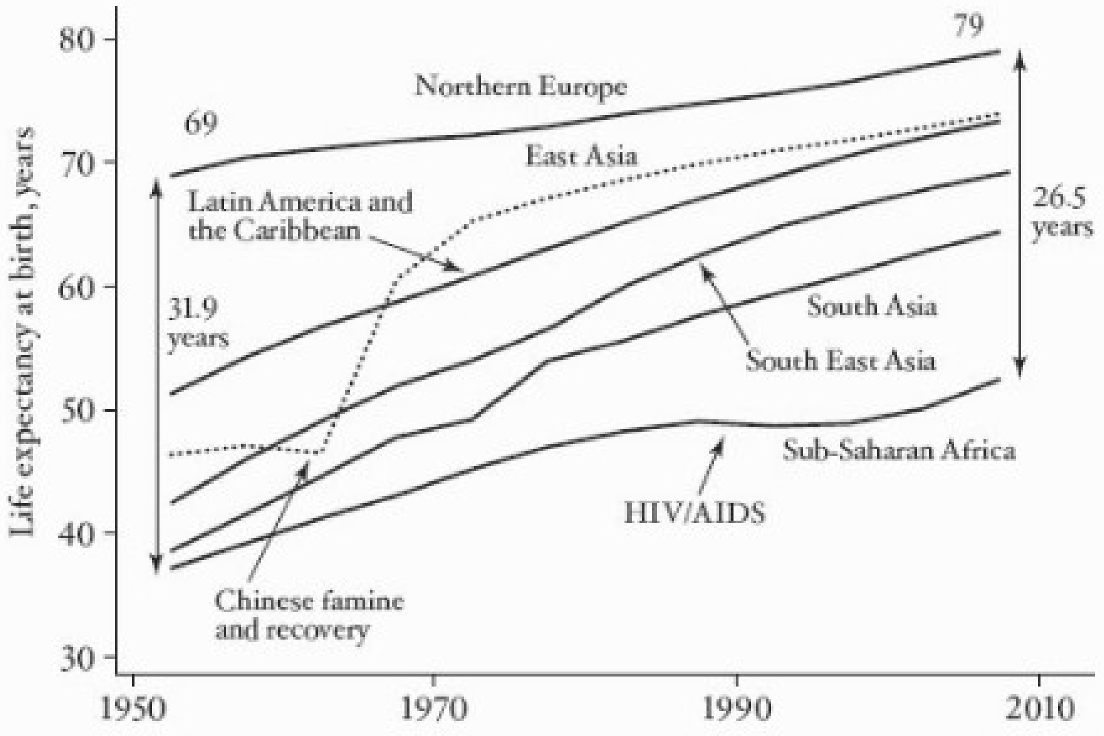


FIGURE 1 Life expectancy in regions of the world since 1950.

Africa and to a lesser extent South Asia (which extends as far north as Afghanistan) are the regions where the most remains to be done. Even before the HIV/AIDS epidemic, life expectancy in sub-Saharan Africa was growing more slowly than elsewhere, and HIV/AIDS caused a further stalling that is clearly visible in the figure. With the advent in recent years of antiretroviral therapy, and with behavioral change, the UN estimates that African life expectancy has begun to rise once again. Yet in the most affected countries most or all of the postwar progress was lost; life expectancy in Botswana—one of the best-governed and economically successful countries in Africa—rose from 48 years to 64 years and then fell back to 49 years in 2000–05, while Zimbabwe's life expectancy—in one of the worst-governed and economically unsuccessful

countries in Africa—was lower in 2005–10 than in 1950–55. Great epidemics that kill millions of people—according to WHO, HIV/AIDS had killed thirty-four million by the end of 2011—clearly did not end after the influenza epidemic of 1918–19, nor should we be complacent about the absence of new epidemics in the future.

GLOBAL MORTALITY IN 2008, AND IN THE POOREST AND RICHEST COUNTRIES

	World	Low-income	High-income
Percentages of deaths (percentages of population)			
Ages 0–4	14.6 (9)	35.0 (15)	0.9 (6)
Ages 60 and above	55.5 (11)	27.0 (6)	83.8 (21)
Cancer	13.3	5.1	26.5
Cardiovascular disease	30.5	15.8	36.5
Millions of deaths			
Respiratory infections	3.53	1.07	0.35
Perinatal deaths	1.78	0.73	0.02
Diarrheal disease	2.60	0.80	0.04
HIV/AIDS	2.46	0.76	0.02
Tuberculosis	1.34	0.40	0.01
Malaria	0.82	0.48	0.00
Childhood diseases	0.45	0.12	0.00
Nutritional deficiencies	0.42	0.17	0.02
Maternal mortality	0.36	0.16	0.00
From all causes	56.89	9.07	9.29
Total population	6,737	826	1,077

SOURCE: World Health Organization, Global Health Observatory Data Repository, downloaded February 3, 2013.

NOTES: Cardiovascular disease includes stroke. Respiratory infections are mostly lower respiratory infections (*lower* refers to infections below the vocal chords, including pneumonia, bronchitis, and influenza, which can also affect the upper respiratory tract). Perinatal deaths are deaths of children at birth or immediately thereafter and include deaths associated with babies being premature and of low birth weight, babies who die during birth, and babies who die from infections immediately after birth. Childhood diseases are whooping cough, diphtheria, polio, measles, and tetanus. About two-thirds of deaths from nutritional deficiencies are due to deficiency of protein or energy, and one-third are due to anemia.

The major killers in poor countries are largely the same diseases that used to kill children in the now-rich countries—lower respiratory infections, diarrhea, tuberculosis, and what WHO calls "childhood diseases": whooping cough, diphtheria, polio, measles, and tetanus; between them, these four categories still cause nearly eight million deaths a year. Other important causes of death are malaria and HIV/AIDS (for which treatment is still far from perfect), deaths at or near birth (perinatal deaths), deaths of mothers associated with

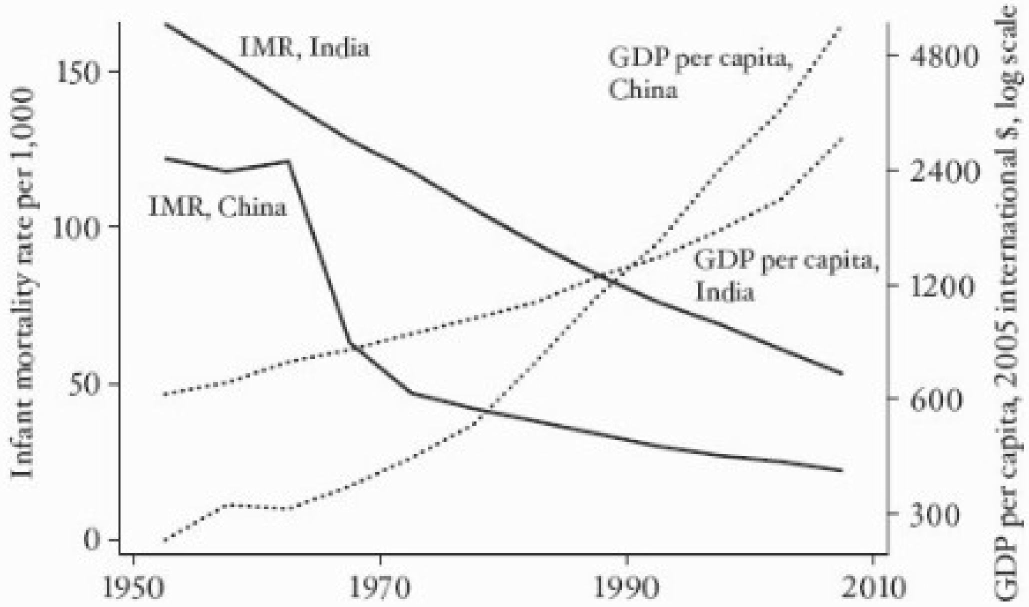
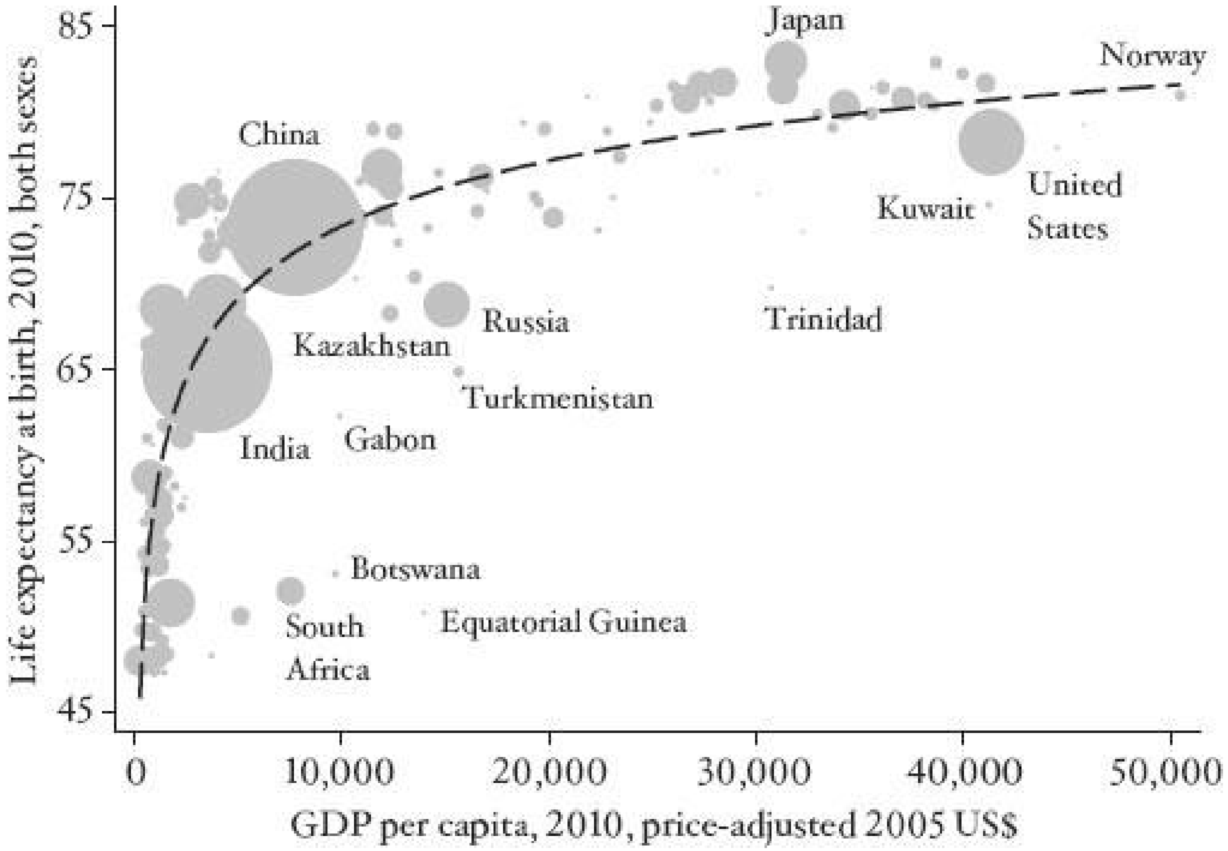


FIGURE 2 Infant mortality and economic growth in China and India.

So we have a puzzle. Why should children die in poor countries when they would not die if they had been born in rich countries? What is it that prevents the knowledge that is freely available and effective in the rich world from saving the lives of millions of people who die in the poor world? The most obvious candidate is poverty. Indeed the very classification I have adopted, between low- and high-income countries, suggests that income is what matters. Just as in the historical context, we think of diarrhea, respiratory disease, tuberculosis, and undernutrition as "diseases of poverty," as we think of cancer, heart disease, and stroke as "diseases of affluence." As was the case in the eigh-

HEALTH AND WEALTH OF NATIONS

LIFE EXPECTANCY AND INCOME



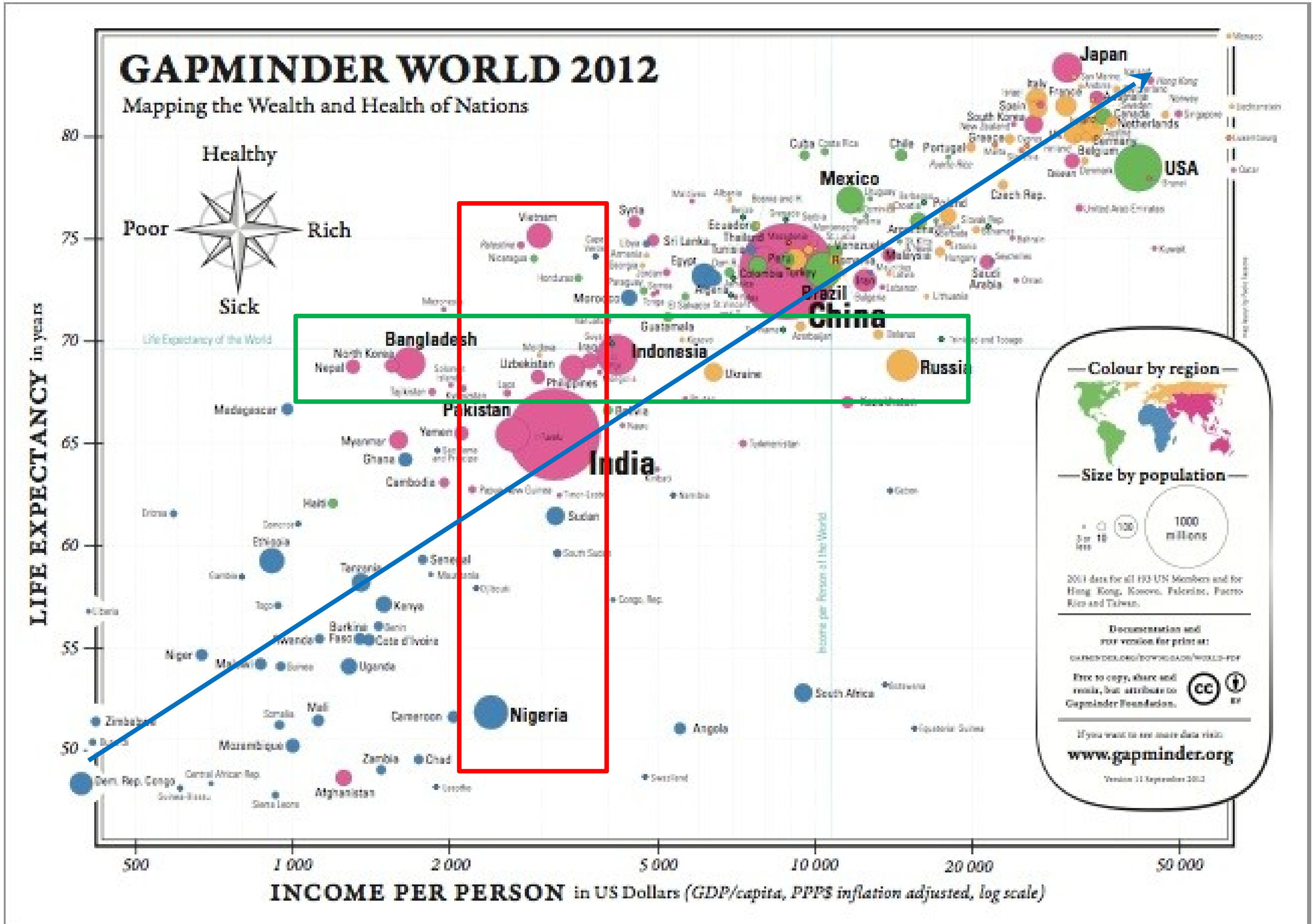
Samuel H. Preston
(1943-)
Professor, Sociology
University of
Pennsylvania

"His empirical account of the changing relationship between a country's life expectancy and its real per capita income —known as the **Preston curve**—*has become the canonical foundation of research on population health."*

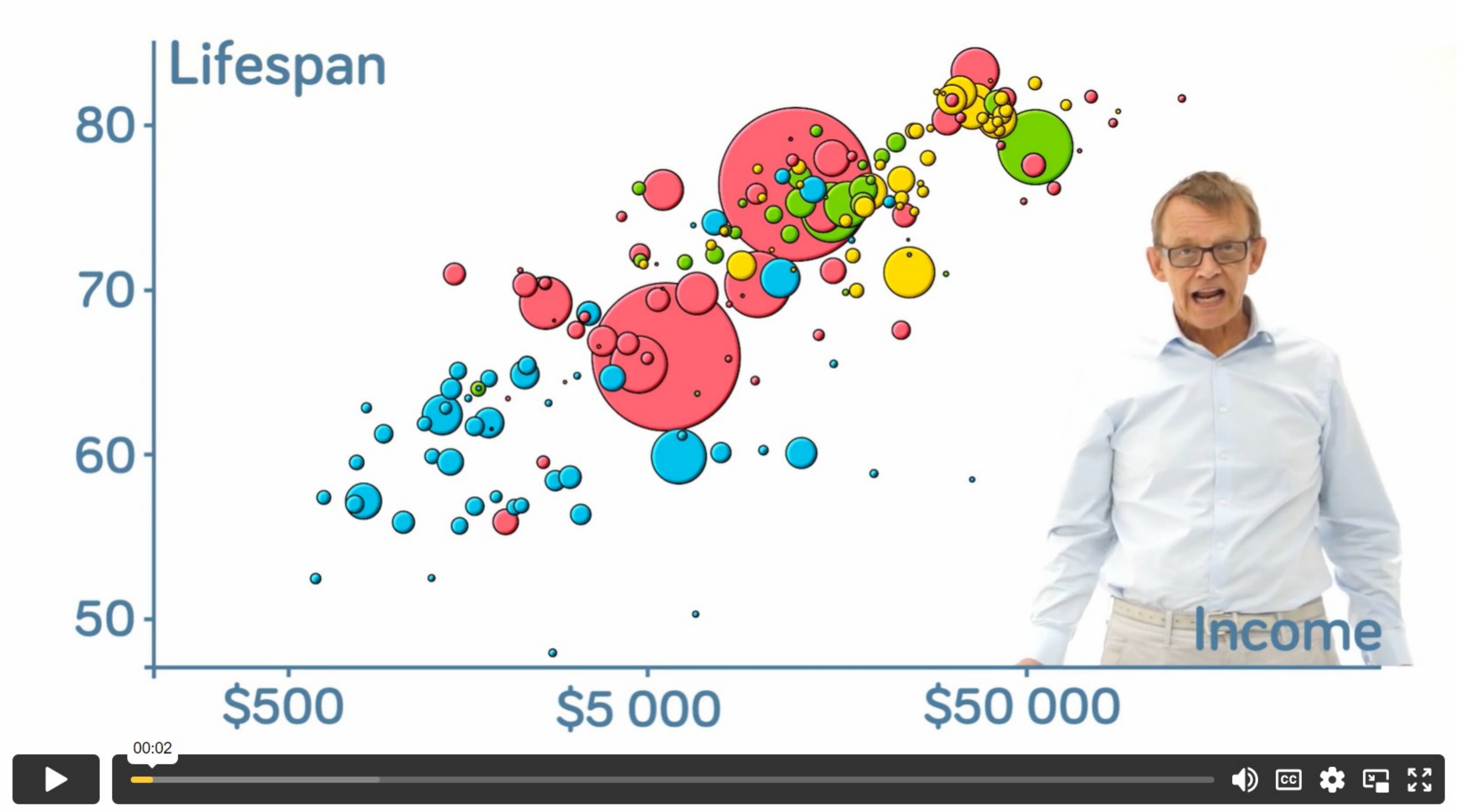
Pioneered by Sam Preston

https://www.populationassociation.org/support/honored-members/samuel-preston

HEALTH AND WEALTH OF NATIONS



Hans Roseling



https://www.gapminder.org/fw/world-health-chart/

IMPROVEMENTS IN HEALTH

- Germ Theory of disease
- Improving sanitation;
- Hygiene, Isolation (quarantine) , Antiseptics
- Vaccines and Antibiotics
- Massive immunization Programs
- Lowcost interventions – ORT for diarrhea